Thomas Robert Butler IV

Senior Software Engineer | AI Engineering & Blockchain
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Technical Skills

Languages: Python, Go, Rust, TypeScript, Solidity, Cairo, SQL, Nix, Bash
Tools & frameworks: PyTorch, Hugging Face, vLLM, SGLang, Kubernetes, Docker, AWS, Linux/NixOS, Temporal, ConnectRPC, PostgreSQL, Redis, Prometheus, Grafana, Foundry, Tree-sitter, React/React Native

Professional Experience

Senior Software Engineer — GFX Labs
September 2025 - Present

- Developed Oku's perpetuals trading backend in Go, connecting real-time market data to **durable order execution** through ConnectRPC, Redis Streams, and Temporal.
- Engineered financial workflows for safe retries and recovery, combining signed-action validation, **replay protection**, and on-chain settlement checks with regression testing.
- Traced failures across database, authentication, and blockchain boundaries; resolved concurrency and transaction-simulation bugs and added **regression tests** to prevent recurrence.
- Equipped coding agents with **structured runtime diagnostics** for inspecting application state and failures, using bounded event capture and allowlisted data exposure.
- Delivered React/TypeScript trading interfaces, including Plasma's $2.4B token launch; established shared UI and review practices and mentored **3 junior engineers**.

Senior Software Engineer — Nakamoto LLC
January 2025 - June 2025

- Led two engineers building AI and data infrastructure for Bittensor subnets.
- Fine-tuned **tool-calling LLMs**, evaluated against the Berkeley Function Calling Leaderboard benchmark, and achieved **top 10% subnet performance**.
- Investigated **subnet evaluation weaknesses** and exploited scoring flaws to win Bittensor subnet competitions.
- Unified datacenter and cloud servers in a Kubernetes cluster for **autoscaled AI inference**; used USB-to-PCIe GPU connections to work around server-rack space constraints.

Mobile Software Engineer — Tholos
October 2023 - January 2025

- Owned iOS/Android applications for an MPC wallet securing over **$35 million** in assets; implemented WebSocket and AWS secure enclave integrations.
- Built **distributed key backup and recovery** that handled network failures, device disconnects, and interrupted operations.
- Implemented key management, signing, and integrations across Ethereum, EVM L2s, Bitcoin, Solana, and Cosmos.

Founder — Suffix Labs
July 2023 - Present

- Founded and operate a software consultancy serving clients across Web3 and crypto lending.
- Delivered client projects spanning Solidity smart contracts and React/TypeScript applications, collaborating with remote teams on implementation, testing, and integration.

Cloud Software Engineer — Maverick Software Consulting -- Thomson Reuters
June 2021 - May 2023

- Built an AWS resource-cleanup application in **Python**, Lambda, and Step Functions with projected initial savings of $14 million.
- Owned unit/integration tests, CI/CD, and documentation; contributed to an RDS tool used across cloud teams.

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Selected Projects

Argos — LLM Development Infrastructure · Source
- Built a Rust CLI/TUI for **parallel LLM-assisted development** with task-specific NixOS microVMs and local/remote tmux sessions over SSH.
- Built cancellable discovery and subprocess supervision with deadlines, output limits, and process-group cleanup; verified lifecycle behavior with **Python integration tests**.

ML-Based Smart Contract Vulnerability Detection — University Project · Source
- Applied **Python**/**PyTorch** to **smart contract vulnerability detection**, extracting opcode features and converting bytecode into RGB images for neural-network classification experiments.

Multi-Machine Nix Infrastructure
- Maintain declarative configurations for **7 Linux/macOS hosts** through Nix flakes, NixOS/nix-darwin, and shared Home Manager modules.
- Built reproducible **Python**, Rust, Go, and TypeScript environments and declarative k3s configuration.

Starknet Validator Deployment & Monitoring · Source
- Built Kubernetes deployment tooling for a Pathfinder full node and two validators, with Prometheus alerts, Grafana dashboards, and automated operational reports.

ZKGenomics — Proof Generation & AI Backend · Source
- Built a Go prototype for gnark/Groth16 proof generation and verification, including VCF parsing, circuit/witness construction, and serialization.
- Integrated an LLM API with proof-result context, user knowledge files, and PostgreSQL persistence.

Open Source Experience

Contributor — Kakarot zkEVM · Source
September 2023
- Implemented EVM SHL execution and checked/wrapping bit shifts in Cairo, with **boundary-case tests**; contributions **merged upstream**.

Maintainer — Starknet Phone · Source
September 2024 - Present (development paused)
- Ported the Beerus Starknet light client to Android and implemented Java bindings in Rust using Android NDK/JNI; maintained the repository and supported **15+ contributors**.

Maintainer — Cairo Zed Extension · Source
July 2024 - Present
- Developed and maintain Cairo support for Zed with **55,000+ downloads**, using Rust, Tree-sitter, and LSP; updated grammar definitions and parser corpus tests.

Contributor — OpenCode · Source
April 2026
- Fixed a missing dependency breaking Nix builds; package, lockfile, and build-hash updates **merged upstream**.

Education

University of Iowa — BSE Computer Science and Engineering — Focus Area: **Machine Learning**